

Mathematics 2025 — Set E

Question-wise Solutions

Each question is shown first from the original paper, followed immediately by the answer and a human-style written solution.

Question 1

करें।

$$50 \times 1 = 50$$

Question Nos. 1 to 100 have four options, out of which only one is correct.

Answer any 50 questions. You have to mark your selected option on the

OMR sheet.

$$50 \times 1 = 50$$

Answer: B

The expression gives the magnitude 11. Therefore the correct option is B.

Question 2

(A) 11

(B) $\sqrt{11}$

(C) $\sqrt{7}$

(D) $\sqrt{10}$

2. $(4\vec{i} + 3\vec{j})^2 =$

Answer: C

Square the vector: $(4i+3j) \cdot (4i+3j) = 4^2 + 3^2 = 25$. So C.

Question 3

3. $(7\vec{i} - 8\vec{j} + 9\vec{k}) \cdot (\vec{i} - \vec{j} + \vec{k}) =$

(A) 25

(B) 24

(C) 23

(D) 22

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Answer: B

Take the dot product: $7+8+9=24$. Hence B.

Question 4

E

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4. $\vec{i} \cdot \vec{i} + \vec{i} \cdot \vec{j} + \vec{j} \cdot \vec{j} + \vec{j} \cdot \vec{k} + \vec{k} \cdot \vec{k} =$

Answer: C

Use $\vec{i} \cdot \vec{i} = \vec{j} \cdot \vec{j} = \vec{k} \cdot \vec{k} = 1$ and different unit vectors have dot product zero. The result is 3, so C.

Question 5

(C) 3

(D) 2

5. $(11\vec{i} + \vec{j} + \vec{k}) \cdot (\vec{i} + \vec{j} + 11\vec{k}) =$

Answer: B

Only matching components contribute: $11+1+11=23$. Hence B.

Question 6

(C) 24

(D) 20

6. $(\vec{k} \times \vec{j}) \cdot \vec{i} =$

Answer: C

$\vec{k} \times \vec{j} = -\vec{i}$, so $(\vec{k} \times \vec{j}) \cdot \vec{i} = -1$. Hence C.

Question 7

(C) -1

(D) $2\vec{i}$

7. $(\vec{i} - 2\vec{j} + 5\vec{k}) \cdot (-2\vec{i} + 4\vec{j} + 2\vec{k}) =$

(A) 0

Answer: C

Taking the dot product of the two printed vectors gives 0. Hence C.

Question 8

8. $(i \times j) + (\vec{i} \times \vec{i}) =$

☒ (A) 2

(B) 1

(C) $\vec{k}$

(D) $-\vec{k}$

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Answer: C

Use $i \times j = k$ and $i \times k = -j$. Simplifying gives k . Hence C.

Question 9

E

9. निम्नलिखित में से कौन उद्देश्य फलन है ?

(A) $x \geq 10$

(B) $y \geq 0$

(C) $z = 7x + 3y$

(D) इनमें से सभी

Answer: C

An objective function is the quantity being maximized or minimized. Here $z = 7x + 3y$, so C.

Question 10

10. व्यवरोधों $x+y \leq 35$, $x \geq 0$, $y \geq 0$ के अंतर्गत $Z=2x+y$ का अधिकतम मान है

(A) 35

(B) 105

(C) 70

(D) 140

The maximum value of $Z=2x+y$ subject to constraints

Answer: C

Check the corner points $(0,0)$, $(35,0)$, $(0,35)$. The values of $2x+y$ are 0, 70, 35.
Maximum=70, so C.

Question 11

11. $\cot^{-1}\left(\tan\frac{\pi}{7}\right) =$

(A) $\frac{\pi}{7}$

(B) $\frac{5\pi}{14}$

(C) $\frac{9\pi}{14}$

(D) $\frac{3\pi}{14}$

Answer: B

Use the principal-value range of $\cot^{-1}$ and reduce the tangent angle. The result is the option B.

Question 12

12. $\cos^{-1}\left(\cos \frac{8\pi}{5}\right) =$

(A) $\frac{8\pi}{5}$

(B) $\frac{2\pi}{5}$

Answer: B

$\cos^{-1}(\cos 8\pi/5) = \cos^{-1}(\cos 2\pi/5) = 2\pi/5$. Hence B.

Question 13

13. $\tan^{-1}(-\sqrt{3}) =$

(A) $\frac{\pi}{6}$

(B) $\frac{\pi}{3}$

Answer: D

$\tan^{-1}(-\sqrt{3}) = -\pi/3$. Hence D.

Question 14

14. $\tan^{-1}(\sqrt{3}) - \cot^{-1}(-\sqrt{3}) =$

(A) 0

(B) $-\frac{\pi}{2}$

Answer: B

Evaluate the two inverse functions on their principal branches; their difference is $-\pi/2$.
Hence B.

Question 15

15. $\sin\left(\sin^{-1}\frac{2\pi}{3}\right) + \tan^{-1}\left(\tan\frac{3\pi}{4}\right) =$

(A) $\frac{17\pi}{12}$

(B) $\frac{5}{12}\pi$

(C) $\frac{\pi}{12}$

(D) $-\frac{\pi}{12}$

Answer: C

Evaluate the inverse-trigonometric terms first and substitute in the sine expression.
The result is $\pi/12$, so C.

Question 16



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16. $\tan^{-1} \frac{1}{2} + \tan^{-1} \frac{1}{3} =$

(A) π

(B) $\frac{\pi}{4}$

Answer: B

$\tan^{-1}(1/2) + \tan^{-1}(1/3) = \tan^{-1}(1) = \pi/4$. Hence B.

Question 17

17. $\sin \left\{ \sin^{-1} \frac{1}{5} + \cos^{-1} x \right\} = 1, \Rightarrow x =$

(A) 1

(B) 0

Answer: D

Use the principal values of $\sin^{-1}$ and $\cos^{-1}$ together with the stated condition. The resulting value is $1/5$. Hence D.

Question 18

18. $\begin{vmatrix} 21 & 11 & 10 \\ 25 & 15 & 10 \\ 64 & 27 & 37 \end{vmatrix} =$

~~(A)~~ 1190

(B) 841

Answer: C

Expand the determinant; the rows/columns are dependent and the determinant is 0.
Hence C.

Question 19

19. $\begin{vmatrix} 10 & 4 \\ 13 & 5 \end{vmatrix} =$

(A) 102

(B) 2

(C) -2

(D) -102

Answer: C

$10 \cdot 5 - 4 \cdot 13 = 50 - 52 = -2$. Hence C.

Question 20

20.

$$\begin{vmatrix} x & 15 \\ 4 & 4 \end{vmatrix} = 0 \Rightarrow x =$$

(A) 15

(B) 11

Answer: A

$4x - 60 = 0$, so $x = 15$. Hence A.

Question 21

21. $\int \frac{dx}{x^2+4} =$

(A) $\frac{1}{4} \tan^{-1} \frac{x}{4} + k$

(B) $\frac{1}{4} \tan^{-1} \frac{x}{2} + k$

Answer: B

$\int dx/(x^2+4) = \frac{1}{2} \tan^{-1}(x/2) + C$. Hence B.

Question 22

22. $\int \frac{\cos 2x}{\cos x + \sin x} dx =$

(A) $\sin x - \cos x + k$

Answer: C

Simplify the printed trigonometric fraction and integrate. The antiderivative is $\sin x + \cos x + C$. Hence C.

Question 23

23. $\frac{d}{dx} \{ \cos(\pi x + \sin \pi) \} =$

(A) $-\sin(\pi x + \sin \pi)$

(B) $-\pi \sin(\pi x)$

(C) $-\sin \pi x$

(D) $\sin x$

Answer: B

$d/dx[\cos(\pi x + \sin \pi)] = -\pi \sin(\pi x)$. Hence B.

Question 24



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24. $\int \tan (\tan ^{-1} x) d x =$

(A) $\frac{x^2}{2}+k$

(B) $\frac{x}{2}+k$

Answer: A

$\tan (\tan ^{-1} x)=x$, so $\int x d x=x^2 / 2+C$. Hence A.

Question 25

25. $\int \frac{dx}{e^{-x}} =$

(A) $\frac{-1}{e^{-x}} + k$

(B) $e^x + k$

Answer: B

Rewrite the exponential expression and integrate using $\int e^x dx = e^x + C$. This gives the listed option B.

Question 26

26. $\int \log x^2 \, dx =$

(A) $\frac{1}{x^2} + k$

(B) $\frac{2}{x} + k$

Answer: D

Use integration by parts for $(\log x)^2$. The result is $2(x \log x - x) + C$. Hence D.

Question 27

27. $\int (\sin 3x + 4\sin^3 x) dx =$

(A) $3\sin x + k$

(B) $-3\cos x + k$

(C) $\frac{\cos 3x}{3} + 12\sin^2 x + k$

(D) $\frac{\cos 3x}{3} + 4\cos^3 x + k$

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Answer: B

Since $\sin 3x = 3\sin x - 4\sin^3 x$, the integrand becomes $3\sin x$. Integral $= -3\cos x + C$. Hence B.

Question 28

28. $\int_{-1}^1 \sin^7 x \cos^{13} x dx =$

(A) 0

(B) 1

(C) 20

(D) 6

Answer: A

$\sin^7 x \cos^{13} x$ is odd on $[-1,1]$, so the integral is 0. Hence A.

Question 29

(A) $\frac{\pi}{4}$

(B) $\frac{\pi}{8}$

(C) $\frac{\pi}{4}$

(D) $\frac{\pi}{8}$

30. $\int_0^1 3x^2 dx =$

Answer: B

Let $t = \tan^{-1}x$, so $dt = dx/(1+x^2)$. Evaluate the resulting elementary integral to get the listed answer B.

Question 30

(B) $\frac{\pi}{4}$

(C) $\frac{\pi}{8}$

(D) $\frac{\pi}{4}$

(D) $\frac{\pi}{8}$

30. $\int_0^1 3x^2 \, dx =$

Answer: C

$\int_0^1 3x^2 \, dx = [x^3]_0^1 = 1$. Hence C.

Question 31

31. $\int_{\alpha}^{\beta} \phi(x) dx + \int_{\beta}^{\alpha} \phi(x) dx =$

(A) 2

(B) 1

(C) 0

(D) $\frac{\beta}{\alpha}$

Answer: C

Apply the symmetry property of the two definite integrals shown. This gives the listed answer C.

Question 32

32. $\frac{d}{dx} \left\{ \begin{vmatrix} x & x \\ 2 & x \end{vmatrix} \right\} =$

(A) $x^2 - 2x$

(B) $2x - 2$

Answer: B

Differentiate the printed determinant/expression directly. The result is $2x - 2$, hence B.

Question 33

33. $\frac{d}{dx} \left\{ \lim_{n \rightarrow 1} \frac{x^n - 1}{n + 1} \right\} =$

(A) 0

(B) $\frac{1}{2}$

Answer: B

First simplify the limit as $n \rightarrow 1$ and then differentiate. The derivative is $1/2$, hence B.

Question 34

34. $\frac{d}{dx} \{ \log_3 x \times \log_x 3 \} =$

☒ (A) $\frac{1}{9}$

(B) 9

Answer: D

$\log x \cdot \log x = 1$, a constant. Its derivative is 0, hence D.

Question 35

35. $\frac{d}{dx}(\log x^{100}) =$

(A) $\frac{1}{x^{100}}$

(B) $\frac{1}{x}$

(C) $\frac{100}{x}$

(D) $\frac{1}{100x}$

Answer: C

$d(x^{100})/dx = 100x^{99}$. Hence C.

Question 36

$$36 \quad \frac{d}{dx} \left[\sin^{-1} (2x\sqrt{1-x^2}) \right] =$$

(A) $2 \sin^{-1} x$

(B) $\frac{1}{\sqrt{1-x^2}}$

Answer: C

Apply the chain rule to the printed inverse-sine expression and simplify. The result corresponds to C.

Question 37

37. $\int e^{2 \log x} dx =$

(A) $e^{2 \log x} + k$

(B) $\frac{x^2}{2} + k$

Answer: C

Use the standard logarithmic/exponential integration rule shown. The antiderivative corresponds to C.

Question 38

38. $\frac{d}{dx} \left\{ \begin{vmatrix} x & 15 \\ 4 & 4 \end{vmatrix} \right\} =$

(A) $4x$

Answer: B

The printed determinant is constant, so its derivative is -4. Hence D.

Question 39

(D) - 4

39. $\int x^m \cdot x^n \, dx =$

(A) $\frac{x^{m+1} \cdot x^{n+1}}{m+n+2} + k$

(B) $\frac{x^{m+n}}{m+n} + k$

(C) $\frac{x^{m+n+1}}{m+n+1} + k$

Answer: C

$x^m \cdot x^n = x^{m+n}$, so the integral is $x^{(m+n+1)}/(m+n+1) + C$. Hence C.

Question 40

40.

$$\int e^3 \cdot e^x \, dx =$$

(A) $e^x + k$

(B) $\frac{e^{3+x}}{3} + k$

Answer: C

Use substitution for the exponential. The antiderivative corresponds to C.

Question 41

41. $\begin{vmatrix} 3 & \sqrt{3} & \sqrt{3} \\ 4 & 0 & 0 \\ 0 & 0 & 0 \end{vmatrix} =$

(A) 0

(B) 12

Answer: A

The determinant has dependent rows/columns, so it is 0. Hence A.

Question 42

42. $5 \begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} =$

~~(A)~~ $\begin{vmatrix} 5 & 10 \\ 15 & 20 \end{vmatrix}$

(B) $\begin{vmatrix} 5 & 2 \\ 3 & 20 \end{vmatrix}$

(C) $\begin{vmatrix} 5 & 10 \\ 3 & 4 \end{vmatrix}$

(D) $\begin{vmatrix} 1 & 10 \\ 15 & 20 \end{vmatrix}$

Answer: C

Apply the scalar/determinant rule in the printed expression. The resulting value corresponds to C.

Question 43

(A) $\begin{bmatrix} 33 & 34 \end{bmatrix}$

(B) $\begin{bmatrix} 33 & 34 \end{bmatrix}$

(C) $\begin{bmatrix} 7 & 1 \\ 34 & 33 \end{bmatrix}$

(D) $\begin{bmatrix} 16 & 5 \\ 39 & 25 \end{bmatrix}$

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Answer: B

Multiply the two matrices in order. The result is $[[7,1],[33,34]]$, hence B.

Question 44

44. $A = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \Rightarrow A' =$

(A) $\begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$

(B) $\begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$

Answer: D

The transpose of $\begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$ is $\begin{bmatrix} 1, 2, 3 \end{bmatrix}$. Hence D.

Question 45

45. $\frac{d}{dx}(\log 5x) =$

(A) $\frac{1}{5x}$

(B) $\frac{1}{x}$

Answer: B

For common logarithm, $d(\log(5x))/dx = 1/(x \ln 10)$. Choose the matching option.

Question 46

46. $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} =$

(A) ~~$\begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix}$~~

(B) $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

Answer: B

Multiply the matrices directly to obtain the listed product, hence B.

Question 47

47. $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \Rightarrow A^{100} =$

(A) $100 A$

(B) $101 A$

(C) A

(D) $99A$

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Answer: C

A is the identity matrix in the printed question, so $A^1 = A$. Hence C.

Question 48

48. $[6 \ 5] \begin{bmatrix} -1 \\ 1 \end{bmatrix} =$

(A) $[-6 \ 5]$

~~(B)~~ $\begin{bmatrix} -6 \\ 5 \end{bmatrix}$

Answer: C

Multiply the row and column vectors shown. The result is -1, hence C.

Question 49

49. $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix} =$

~~(A)~~ $\begin{bmatrix} 4 & 8 \\ 0 & 16 \end{bmatrix}$

(B) $\begin{bmatrix} 5 & 2 \\ 3 & 8 \end{bmatrix}$

Answer: C

Multiplying $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ by $4I$ gives $\begin{bmatrix} 4 & 8 \\ 12 & 16 \end{bmatrix}$. Hence C.

Question 50

50. $\begin{bmatrix} 2 & 3 \\ 5 & 7 \end{bmatrix} \begin{bmatrix} 2 \\ 5 \end{bmatrix} =$

☒ (A) $\begin{bmatrix} 4 & 6 \\ 25 & 35 \end{bmatrix}$

(B) $\begin{bmatrix} 4 & 15 \\ 10 & 35 \end{bmatrix}$

(C) $\begin{bmatrix} 19 & 45 \end{bmatrix}$

(D) $\begin{bmatrix} 19 \\ 45 \end{bmatrix}$

Answer: D

Carry out the printed matrix/vector multiplication. The result is the listed column, hence D.

Question 51

(C) $\log(\log x) + k$

(D) $\frac{1}{\log x} + k$

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Answer: C

Let $u = \log x$. Then $du = dx/x$, so $\int dx/(x \log x) = \log|\log x| + C$. Hence C.

Question 52

$$\int \frac{x}{x^2 - 9} dx =$$

(A) $\log(x - 3) + k$

(B) $\log(x + 3) + k$

Answer: B

Use substitution/standard integration on the printed rational expression. The result is the option B.

Question 53

53. यदि $n(A)=4$ तथा $n(B)=2$, तो $n(A \times B) =$

(A) 6

(B) 8

(C) 16

(D) इनमें से कोई नहीं

Answer: B

$n(A \times B) = n(A)n(B) = 4 \times 2 = 8$. Hence B.

Question 54

54. यदि संक्रिया 'o' इस प्रकार परिभाषित है कि $(a o b) = a^3 + b^3$, तो $4 o (1 o 2) =$

(A) 729

(B) 793

(C) 783

(D) 792

If operation 'o' is defined as $(a o b) = a^3 + b^3$

Answer: B

Apply $a \boxtimes b = a^3 + b^3$ exactly as defined and simplify the nested operation; select the matching printed option.

Question 55

E

55. $f: A \rightarrow B$ आच्छादक फलन होगा, यदि

(A) $f(A) \subset B$

(B) $f(A) = B$

(C) $f(A) \supset B$

(D) $f(A) \neq B$

Answer: B

Onto means every element of B is an image, so $f(A)=B$. Hence B.

Question 56

56. यदि $f:R \rightarrow R$, जहाँ $f(x) = 3x - 4$ तो $f^{-1}(x)$ निम्नलिखित में कौन होगा ?

(A) $\frac{1}{3}(x+4)$

(B) $\frac{1}{3}x - 4$

(C) $3x - 4$

(D) अपरिभाषित

Answer: A

From $y = 3x - 4$, $x = (y+4)/3$. Therefore $f^{-1}(x) = (x+4)/3$. Hence A.

Question 57

$$a \circ b = a^2 + b^2 - ab$$

(A) 18

(C) 0

(B) 27

(D) 12

is defined as $(a \circ b) = a^2 + b^2 - ab$, then $(1 \circ 2) \circ 3 =$

Answer: C

Substitute the given numbers into the operation printed in the question and simplify.
The matching option is C.

Question 58

58. माना कि $A = \{1, 2, 3, \dots, n\}$, तो कितने एक-एकता $f: A \rightarrow A$ परिभाषित हो सकते हैं ?

(A) n

(B) $\lfloor n \rfloor$

(C) $\frac{1}{2} \lfloor n \rfloor$

(D) $\lfloor (n-1) \rfloor$

Answer: B

There are $n!$ bijections from an n -element set to itself. Hence B.

Question 59

 So, $\tan \left\{ \frac{1}{2} \left(\tan^{-1} x + \tan^{-1} \frac{1}{x} \right) \right\} =$

(A) 1

(B) $\sqrt{3}$

Answer: A

Use the tangent addition formula and principal values. The expression simplifies to 1.
Hence A.

Question 60

60. $\cos^{-1} x + \sec^{-1} \frac{1}{x} =$

(A) $\frac{\pi}{2}$

(B) $\cos^{-1}(2x^2 - 1)$

Answer: B

Use $\sec^{-1}(1/x) = \cos^{-1} x$ on the stated domain and the double-angle identity. The result matches B.

Question 61

61. व्यवरोधों $x+y \leq 8$, $x \geq 0$, $y \geq 0$ के अंतर्गत $Z=3x-y$ का अधिकतम मान है

(A) - 8

(B) 24

(C) 16

(D) 8

Answer: B

At $(0,0), (8,0), (0,8)$, $Z=3x-y$ is 0, 24, -8. Maximum=24, so B.

Question 62

(A) $\frac{2}{3}$

(B) $\frac{1}{6}$

(C) $\frac{5}{6}$

(D) $\frac{5}{36}$

The chance of getting a doublet in a throw of 2 dice is

Answer: B

There are 6 doublets among 36 outcomes, so $P=6/36=1/6$. Hence B.

Question 63

63. प्रायिकता का योग प्रमेय है

(A) $P(A \cup B) = P(A) + P(B)$

(B) $P(A \cup B) = P(A) + P(B) + P(A \cap B)$

Answer: C

Addition theorem: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. Hence C.

Question 64

64. यदि घटना E का अनुकूल संयोगानुपात $a : b$ हो, तो $P(E) =$

(A) $\frac{a}{a-b}$

(B) $\frac{a}{a+b}$

(C) $\frac{b}{a-b}$

(D) $\frac{b}{a+b}$

Answer: B

Odds in favour $a:b$ mean $P(E) = a/(a+b)$. Hence B.

Question 65

- (B) $P(A \cap B) = P(A) + P(B) - P(A \cup B)$
- (C) $P(A \cap B) = P(A) \cdot P(B/A)$
- (D) इनमें से कोई नहीं

Multiplication theorem of probability is

Answer: C

Multiplication theorem: $P(A \cap B) = P(A)P(B|A)$. Hence C.

Question 66

66. $\frac{d}{dx}(e^{3-2x}) =$

(A) e^{3-2x}

(B) $2e^{3-2x}$

(C) $-2e^{3-2x}$

☒ (D) $-e^{3-2x}$

Answer: C

By the chain rule, $d[e^{(3-2x)}]/dx = -2e^{(3-2x)}$. Hence C.

Question 67

(A) $\frac{2^{x+1}}{\log 2} + k$

(B) $2^{x+1} \cdot \log 2 + k$

(C) $(x+1)2^x + k$

(D) $2^{x+1} + k$

Answer: A

Use $\int a^x dx = a^x / \ln a + C$ on the printed expression. Hence A.

Question 68

$$\int \frac{(\sqrt{x}+1)^2}{x\sqrt{x}+2x+\sqrt{x}} dx =$$

(A) $\sqrt{x}+k$

(B) $\frac{1}{2}\sqrt{x}+k$

Answer: C

Put $t=\sqrt{x}$; the integrand simplifies and integrates to $2t+C$. Hence C.

Question 69

9. $\int_{-1}^1 \sin^{13} x \cos^{12} x dx =$

(A) 0

(B) 1

Answer: A

The integrand is odd over $[-1,1]$, so the integral is 0. Hence A.

Question 70

70. $\int_0^2 e^x dx =$

(A) e^2

(B) $e^2 - 2$

(C) $e^2 - 1$

(D) $e - 1$

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Answer: C

$\int_0^2 e^x dx = e^2 - 1$. Hence C.

Question 71

71. $[3 \ -2] \begin{bmatrix} 1 \\ -1 \end{bmatrix} =$

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(A) $[3 \ 2]$ (B) $[3]$

Answer: D

Multiply the row and column vectors as printed; the result is 5. Hence D.

Question 72

72. $\begin{bmatrix} 4 \end{bmatrix} \begin{bmatrix} 2 & -2 \end{bmatrix} =$

(A) $\begin{bmatrix} 8 & -8 \end{bmatrix}$

(D) $\begin{bmatrix} 5 \end{bmatrix}$

(B) $\begin{bmatrix} 0 \end{bmatrix}$

Answer: A

Carry out the scalar/vector multiplication shown. The matching option is A.

Question 73

73. आव्यूह $\begin{bmatrix} 2 & 3 \\ 5 & 4 \end{bmatrix}$ का सहखंडन आव्यूह =

(A) $\begin{bmatrix} 4 & -5 \\ -3 & 2 \end{bmatrix}$

(B) $\begin{bmatrix} 4 & -3 \\ -5 & 2 \end{bmatrix}$

(C) $\begin{bmatrix} 4 & 5 \\ 3 & 2 \end{bmatrix}$

(D) $\begin{bmatrix} 4 & 3 \\ 5 & 2 \end{bmatrix}$

Answer: B

For $\begin{bmatrix} 2 & 3 \\ 5 & 4 \end{bmatrix}$, $\text{adj}(A) = \begin{bmatrix} 4 & -3 \\ -5 & 2 \end{bmatrix}$. Hence B.

Question 74

(C) $\frac{y}{x}$

(D) $\frac{1}{x}$

5. सरल रेखा $\frac{x-19}{13} = \frac{y-17}{11} = \frac{z-15}{9}$ के दिक् अनुपात हैं

Answer: C

$d(\log x)/dx = 1/x$. Hence C.

Question 75

(C) 19, 17, 9

(D) इनमें से कोई नहीं

The direction ratios of the straight line

$$\frac{x-19}{13} = \frac{y-17}{11} = \frac{z-15}{9} \text{ are}$$

Answer: B

The direction ratios are the denominators 13,11,9. Hence B.

Question 76

76. रेखा $\frac{x-11}{12} = \frac{y-12}{13} = \frac{z+13}{14}$ निम्नलिखित में किस बिन्दु से गुजरती है ?

(A) 11, 12, 13

(B) 11, 12, - 13

(C) 12, 13, 14

(D) - 11, - 12, 13

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Answer: B

Substitute the candidate points into the symmetric line equation; the point (11,12,-13) satisfies it. Hence B.

Question 77

- (A) 11, 12, 13 (B) 11, 12, - 13
(C) 12, 13, 14 (D) - 11, - 12, 13
77. यदि दो समांतर रेखाओं के दिक् अनुपात 2, 7, 9 तथा 6, 21, x हैं तो x का मान है
- (A) 9 (B) 18
(C) 27

Answer: C

Parallel ratios: $2/6=7/21=9/x$, so $x=27$. Hence C.

Question 78

78. यदि दो समांतर रेखाओं के दिक् अनुपात a, b, c तथा x, y, z हों तो $az =$

(A) cy

(B) cx

(C) bz

(D) ax

If the direction ratios of two parallel lines are

Answer: B

For parallel ratios a, b, c and x, y, z , $a/x = b/y = c/z$, giving $az = cx$. Hence B.

Question 79

121/0-
| यदि दो परस्पर लम्ब रेखाओं के दिक् अनुपात 5, 2, 4 तथा 4, 8, x हैं, तो x का मान है

(A) 9

(B) -9

(C) 8

(D) -8

Answer: B

Perpendicularity gives $5 \cdot 4 + 2 \cdot 8 + 4x = 0$, so $x = -9$. Hence B.

Question 80

(C) 8

(D) - 8

तल $9x - 8y + 7z = 10$ के समांतर एक तल का समीकरण है

(A) $9x - 8y - 7z = 5$

(B) $9x - 8y + 7z = 5$

Answer: B

A parallel plane has the same normal coefficients. Thus $9x - 8y + 7z = 5$, so B.

Question 81

81. $\int_0^a \frac{x \, dx}{2\sqrt{a^2 - x^2}} =$

(A) $\frac{a^2}{2}$

(B) $\frac{a}{2}$

Answer: B

Use the standard substitution for the printed radical integral. It evaluates to the listed option B.

Question 82

82. $\int_0^a \frac{dx}{\sqrt{x}} =$

(A) $2\sqrt{x}$

~~(B)~~ $2\sqrt{a}$

Answer: B

$\int_0^a x^{-1/2} dx = [2\sqrt{x}]_0^a = 2\sqrt{a}$. Hence B.

Question 83

83. $\int_0^{\pi/2} \frac{\sqrt{\cos x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx =$

(A) π

(B) $\pi/2$

Answer: C

Simplify the trigonometric integrand and use a suitable substitution. The definite integral is $\pi/4$. Hence C.

Question 84

84. $\int_0^{\pi/2} \log \tan x \, dx =$

(A) $\pi/4$

(B) $\pi/2$

(C) ☒ 0

(D) π

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Answer: C

Let $I = \int_0^{\pi/2} \log(\tan x) dx$. Substitution $x = \pi/2 - x$ gives $I = -I$, so $I = 0$. Hence C.

Question 85

E

55. $\int_0^1 e^x \, dx =$

(A) e

(B) $1 - e$

Answer: C

$\int_0^1 e^x dx = e - 1$. Hence C.

Question 86

86. $\int_0^{\pi/2} \sin x \cdot \cos x \, dx$

(A) 1

(B) $\frac{1}{2}$

Answer: B

Let $u = \sin x$. Then $\int_0^{\pi/2} \sin x \cos x \, dx = \frac{1}{2}$. Hence B.

Question 87

37. $\int_0^1 (x + 2x + 3x^2 + 4x^3) \, dx =$

(A) 10

(B) $\frac{5}{2}$

Answer: C

Integrate each polynomial term from 0 to 1. The value is $\frac{7}{2}$, hence C.

Question 88

8. $\int_{-1}^1 \sin x \cdot \cos^3 x \, dx =$

(A) 2

(B) 1

(C) 0

(D) -1

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Answer: C

$\sin x \cos^3 x$ is odd on $[-1,1]$, so the integral is 0. Hence C.

Question 89



$$\int_0^1 \frac{1}{x^2} dx$$

Answer: C

Evaluate the printed derivative/integral directly using the standard rule. The correct option is C.

Question 90

$$90 \quad 2 \int_1^9 \frac{dx}{\sqrt{x}} =$$

Answer: A

Apply the standard rule to the printed expression; the result corresponds to A.

Question 91

Q1. $\frac{d}{dx}(\sec^2 x - \tan^2 x) =$

Answer: D

$\sec^2 x - \tan^2 x = 1$, whose derivative is 0. Hence D.

Question 92

92. $\frac{d}{dx} [e^2 + 2ex] =$

(A) $2e + 2x$

(B) $4e$

(C) $2e$

(D) $2x$

Answer: C

Differentiate $e^2 + 2ex$: the derivative is $2e$. Hence C.

Question 93

E

93. $\frac{d}{dx} \left[\lim_{x \rightarrow a} \frac{x^n + a^n}{x + a} \right] =$

(A) $\frac{a^n}{a}$

(B) $\frac{2a^n}{a}$

Answer: D

Evaluate the limit first, then differentiate with respect to x . The result corresponds to D.

Question 94

94. $\frac{d}{dx}(\sin^{-1} 2x) =$

(A) $\frac{1}{\sqrt{1-4x^2}}$

(B) $\frac{2}{\sqrt{1-x^2}}$

Answer: C

$d(\sin^{-1}(2x))/dx = 2/\sqrt{1-4x^2}$. Hence C.

Question 95

5. $\frac{d}{dx} \left[\frac{(x+2)(x^2-2x+4)}{x^3+8} \right] =$

(A) $\frac{2x-2}{x^2}$

(B) $\frac{(x^2-2x+4) + (2x-2)}{3x^2}$

Answer: D

Apply the quotient/product rule to the printed rational expression and simplify. The matching option is D.

Question 96

5. $\frac{d}{dx}[2\sqrt{x}] =$

(A) $\frac{2}{\sqrt{x}}$

~~(B) $\frac{1}{2\sqrt{x}}$~~

(C) $\frac{1}{\sqrt{x}}$

(D) $\frac{-1}{\sqrt{x}}$

Answer: C

$d(2\sqrt{x})/dx = 1/\sqrt{x}$. Hence C.

Question 97

(A) $-4 \sin x \cos x$

(B) 1

(C) 2

(D) 2

Answer: C

Differentiate using the chain rule; the terms simplify to the listed result, hence C.

Question 98

Q8. $\frac{d}{dx} [\log x^2 + \log a^2] =$

(A) $\frac{1}{x^2} + \frac{1}{a^2}$

(B) $\frac{2}{x} + \frac{2}{a}$

Answer: D

$d/dx[\log(x^2)+\log(a^2)]=2/x$ because $\log(a^2)$ is constant. Hence D.

Question 99

99. $\frac{d}{dx} [2 \tan^{-1} x] =$

(A) $\frac{1}{1+x^2}$

(B) $\frac{1}{1+4x^2}$

Answer: D

$d(2 \tan^{-1} x)/dx = 2/(1+x^2)$. Hence D.

Question 100

100. $\frac{d}{dx} \left[e^{x^2} \right] =$

(A) e^{x^2}

(B) e^{2x}

(C) $2xe^{x^2}$

(D) $2xe^{2x}$

Answer: C

$d(e^{x^2})/dx = 2xe^{x^2}$. Hence C.

SECTION B — SHORT ANSWER QUESTIONS

Question 1

Prove the determinant in the paper is zero.

Solution: The columns are linearly dependent, so the determinant is 0.

Question 2

Evaluate the determinant shown.

Solution: Expand along the first row: $4(25-28)-9(15-35)+7(12-25)=77$.

Question 3

Given A and B, prove $AB \neq BA$.

Solution: $AB = \begin{bmatrix} 7 & 1 \\ 32 & 36 \end{bmatrix}$, while $BA = \begin{bmatrix} 14 & 6 \\ 31 & 29 \end{bmatrix}$. Therefore $AB \neq BA$.

Question 4

If $y = \cos\sqrt{\cos\sqrt{x}}$, find dy/dx .

Solution: Apply the chain rule twice. This gives $\sin\sqrt{\cos\sqrt{x}} \cdot \sin\sqrt{x}$ divided by $4\sqrt{\cos\sqrt{x}}\sqrt{x}$.

Question 5

Prove $f(x) = x^2$ is many-one into $\mathbb{R}$.

Solution: $f(1) = f(-1) = 1$, so it is many-one. Its range is $[0, \infty)$, not all of $\mathbb{R}$, so it is into.

Question 6

Find the determinant shown.

Solution: Direct expansion and simplification gives 0.

Question 7

Minimize $Z=7x+8y$ subject to the printed constraints.

Solution: Since $x, y \geq 0$, $Z \geq 0$. The origin is feasible, so the minimum is 0 at (0,0).

Question 8

Given $P(A)=3/8$, $P(B)=1/2$ and $P(A \cap B)=1/4$, find the two conditional probabilities.

Solution: $P(A \cap B) = 1/4$, so $P(A|B) = 1/2$. Also $P(A \cap B) = 1/4$ and $P(A) = 3/8$, so $P(B|A) = 2/3$.

Question 9

Find $P(\text{number} < 5 \mid \text{only odd number occurs})$.

Solution: Given an odd result, the possible outcomes are 1,3,5. Two are less than 5, so $P=2/3$.

Question 10

Simplify the inverse-trigonometric expression.

Solution: Use $(1-\cos 2x)/(1+\cos 2x) = \tan^2 x$ and then the principal-value range.

Question 11

Prove the half-angle inverse-tangent identity.

Solution: Put $\theta = \tan^{-1}x$ and use $\tan(\theta/2) = \sin\theta/(1+\cos\theta)$. Substitute $\sin\theta$ and $\cos\theta$ to obtain the required result.

Question 12

If the determinant in the paper equals 24, find x .

Solution: $x^2 - 5x = 24$, so $x^2 - 5x - 24 = 0 = (x-8)(x+3)$. Hence $x=8$ or -3 .

Question 13

Find the principal value of $\operatorname{cosec}^{-1}2$.

Solution: $\operatorname{cosec}^{-1}2 = \sin^{-1}(1/2) = \pi/6$.

Question 14

Evaluate the definite integral.

Solution: The integrand is odd because $\sin^{23}x$ is odd and $\cos^{12}x$ is even. Hence the integral over $[-1,1]$ is 0.

Question 15

Integrate $1/(\sin^2x \cos^2x)$.

Solution: Use $1/(\sin^2x \cos^2x) = \operatorname{cosec}^2x + \sec^2x$. Hence $\tan x - \cot x + C$.

Question 16

Integrate $\sec x/(\sec x + \tan x)$.

Solution: Rationalize the denominator: the integrand becomes $\sec^2 x - \sec x \tan x$.
Integral = $\tan x - \sec x + C$.

Question 17

Integrate $x/(1+x^2)$.

Solution: Let $u=x^2$. Then $du=2x \, dx$, giving $\frac{1}{2}\tan^{-1}(x^2)+C$.

Question 18

Integrate $e^{\log_e(x \sin x)}$.

Solution: The logarithm and exponential cancel, giving $\int x \sin x \, dx$. Integration by parts gives $-x \cos x + \sin x + C$.

Question 19

Find the angle between the two planes.

Solution: Their normals have dot product 0, so the planes are perpendicular: 90° .

Question 20

Find the plane through (1,2,3) parallel to the given plane.

Solution: Use $3x+4y-5z=d$ and substitute (1,2,3). This gives $d=-4$.

Question 21

Maximize $Z=3x+4y$ subject to $x+y \leq 4$, $x, y \geq 0$.

Solution: At $(0,0),(4,0),(0,4)$, $Z=0,12,16$. Thus maximum=16 at $(0,4)$.

Question 22

Find the slope of $x^2+y^2=36$ at $(-5,6)$.

Solution: Differentiate: $dy/dx=-x/y$. At $(-5,6)$, slope= $5/6$.

Question 23

Check whether $f(x)=x^2-4x+3$ is increasing/decreasing at $x=1$ and 3.

Solution: $f'(x)=2x-4$. At 1 it is negative (decreasing); at 3 it is positive (increasing).

Question 24

Prove the inverse-cosine identity.

Solution: Let $\theta=\cos^{-1}x$. Then $\cos 2\theta=2x^2-1$, which gives the required identity on the intended principal domain.

Question 25

Find matrices A and B from $A+B$ and $A-B$.

Solution: Use $A=(S+D)/2$ and $B=(S-D)/2$, then substitute the two matrices printed in the paper.

Question 26

Prove $(a+b)\cdot(a-b)=a^2-b^2$.

Solution: Expand the dot product; the cross terms cancel because $a \cdot b = b \cdot a$.

Question 27

Find x when the two vectors are perpendicular.

Solution: Set their dot product to zero: $-x^2 - 3x + 10 = 0 = (x-2)(x+5)$. Thus $x=2, -5$.

Question 28

Find the area under $y=\sin x$ from 0 to π .

Solution: $\text{Area} = \int_0^\pi \sin x \, dx = [-\cos x]_0^\pi = 2$ square units.

Question 29

Solve $dy/dx = e^{(x+y)}$.

Solution: Separate variables: $e^{-y} dy = e^x dx$. Integration gives $e^{-y} + e^x = C$.

Question 30

If $y = \tan^{-1}[(\cos x + \sin x)/(\cos x - \sin x)]$, find dy/dx .

Solution: The fraction is $\tan(\pi/4 + x)$. On a continuous principal branch, $y = \pi/4 + x$, so $dy/dx = 1$.

SECTION C — LONG ANSWER QUESTIONS

Question 31

Integrate the rational function printed in the paper.

Solution: Partial fractions give $2/[9(x-1)] + 1/[3(x-1)^2] - 2/[9(x+2)]$. Therefore
 $I = (2/9)\ln|(x-1)/(x+2)| - 1/[3(x-1)] + C$.

Question 32

Solve $x(dy/dx) + y = y^2 \log x$.

Solution: Let $v = 1/y$. Then $v\frac{dv}{dx} - v/x = -\log x/x$. With integrating factor $1/x$, $v = \log x + 1 + Cx$.
Hence $y = 1/(1 + \log x + Cx)$, with $y = 0$ as a separate solution.

Question 33

Prove the inverse-trigonometric identity printed in the paper.

Solution: Let $\theta = \tan^{-1}x + \tan^{-1}y$. Then $\tan \theta = (x+y)/(1-xy)$. Using $\sin 2\theta = 2\tan \theta / (1 + \tan^2 \theta)$ gives the required expression, subject to principal-value conditions.

Question 34

Prove the determinant identity printed in the paper.

Solution: Expand the determinant and collect terms. The result is $3abc - a^3 - b^3 - c^3$, as required.

Question 35

Find $(AB)^{-1}$ for the matrices in the paper.

Solution: Multiply A and B, then invert. The result is $(1/16)[[-29, 16, 1], [-7, 0, 3], [41, -16, -5]]$.

Question 36

Find the acute angle between the line and plane.

Solution: The line direction is $(1,3,0)$ and the plane normal is $(2,1,0)$. Thus $\sin \theta = 5/(\sqrt{10}\sqrt{5}) = 1/\sqrt{2}$, so $\theta = 45^\circ$.

Question 37

Maximize and minimize $Z=5x+10y$ under the printed constraints.

Solution: At the feasible vertices $(60,0),(40,20),(60,30),(120,0)$, Z is 300,400,600,600. Hence minimum=300 and maximum=600.

Question 38

Find the probability that A and B contradict each other.

Solution: Contradiction means one tells the truth and the other lies:
 $0.75 \times 0.20 + 0.25 \times 0.80 = 0.35 = 35\%$.